

Reproducibility budgets for ML preprints

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Abstract

We argue that preprint platforms should require authors to declare a reproducibility budget at submission time—an explicit specification of the compute, data, and human-time required to replicate the paper’s central claims. We evaluate adoption costs over 12 months and report mixed results: smaller labs benefit, large labs resist.

1 Claims

Claim 1 (Claim 1). Reproducibility-budget declarations correlate with 2.3x higher replication success rates ($p < 0.05$).

Replication status: replicated.

Claim 2 (Claim 2). Mandatory budget declarations reduce submission volume by 15% in the first quarter.

Replication status: contradicted.

Claim 3 (Claim 3). Budgets self-correct via community feedback over a 6-month window.

Replication status: partial.

2 Notes

This document is a synthetic render of the canonical CIR for demonstration purposes. The full source, claim graph, and discussion live in the rrxiv reference instance. See the rrxiv protocol repository for the schema.